Customer Churn Analysis Report

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1. Introduction

Customer churn, where customers leave a service, is a major concern for the telecom industry. Retaining existing customers is more cost-effective than acquiring new ones.

This project analyzes a telecom customer churn dataset to:

- Understand customer behavior, service usage, demographics, and pricing
- Identify key factors driving churn
- Explore predictive modeling for churn

The analysis uses exploratory data analysis (EDA), visualizations, and a Random Forest machine learning model to generate insights for improving customer retention.

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2. Data Overview

Three datasets were used:

1. **Telecom Customer Churn Data** – customer details, services, churn status
2. **Data Dictionary** – definitions of each column
3. **Zip Code Population Data** – regional population info

Data Cleaning Steps:

- Missing values were filled using forward-fill or appropriate defaults (e.g., 'No Offer', 'Unknown Reason')
- Duplicate rows and customer IDs were checked and removed
- Datasets were merged on `Zip Code`

Observation:

- Dataset contains some imbalance in churn classes (more non-churn than churn customers)

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3. Exploratory Data Analysis (EDA)

3.1 General Insights

- Overall, a significant portion of customers are leaving the service.
- Gender distribution is nearly equal.
- Most churn is due to **competition (46.78%)**, followed by **dissatisfaction (16.91%)** and **attitude (15.60%)**.

3.2 Customer Demographics

- **Age:** Highest churn in customers aged >50.
- **Marital Status:** Unmarried customers show higher churn.
- **Dependents:** Fewer dependents correlate with higher churn.

3.3 Service & Usage

- **Contract Type:** Month-to-month contracts have the highest churn.
- **Monthly Charges:** Higher monthly charges are linked to higher churn.
- **Tenure:** Customers with shorter tenure leave more frequently.
- **Offers:** Customers without offers churn more often.
- **Value-added Services:** Customers without online security, backup, device protection, tech support, and streaming services have higher churn.
- **Phone & Internet Services:**
  - Customers without phone service churn more
  - Customers without internet service churn more compared to Fiber Optic or DSL users

3.4 Regional Analysis

- Customers in **low-population areas** have higher churn rates.

3.5 Payment Method

- Electronic check users have higher churn than other payment methods.

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4. Correlation Analysis

- Weak correlation between age, tenure, and monthly charges
- Age has minimal impact on churn, while **contract type, tenure, and monthly charges** are the strongest predictors

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5. Machine Learning Model

Model Used: Random Forest Classifier

Target Variable:

- Churn = 1 (if customer churned)
- Churn = 0 (if customer did not churn / unknown)

Results:

- High accuracy observed, but inflated due to **class imbalance**
- Confusion matrix shows predictions largely on the majority class
- Oversampling (SMOTE) was attempted but did not improve performance due to insufficient minority class samples

Conclusion: Model demonstrates workflow but is **limited by data imbalance**. It is suitable for educational purposes rather than deployment.

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6. Key Findings

1. **Churn Drivers:**
   - Month-to-month contracts
   - Higher monthly charges
   - Short customer tenure
   - Lack of value-added services
   - No active offers
2. **Demographic Patterns:**
   - Older customers (>50)
   - Unmarried with fewer dependents
3. **Service Usage & Payment:**
   - Customers without phone or internet services
   - Electronic check users
4. **Regional Impact:**
   - Higher churn in low-population areas

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7. Limitations

- Severe class imbalance affects ML predictions
- SMOTE and balancing techniques could not fully resolve prediction bias
- Some features like age show minimal correlation with churn

8. Conclusion & Recommendations

Conclusion:
The analysis confirms that churn is mainly influenced by contract type, monthly charges, tenure, and service subscriptions. Customers with shorter tenure, higher charges, or fewer bundled services are at higher risk of leaving.

Recommendations:

1. Promote long-term contracts to reduce churn
2. Offer incentives or bundled services to retain customers
3. Target high-risk demographics (older, unmarried, low-population areas) with retention strategies
4. Improve service offerings to enhance customer satisfaction

Machine Learning Insight:
Random Forest model demonstrates the workflow and identifies key predictors, but results are affected by class imbalance. Future work can include larger, more balanced datasets for better predictive performance.

Project Complete – Mehwish Azam